Environment benchmark: masked correctness and factual consistency

Percent of all questions in the set each row was scored on, by the judge panel's voted label. Configurations 1–7: masked correctness on set A (141 questions with a value hidden from the case), factual consistency on set B (272 questions the case never answers). Configuration 8: both on set C (272 questions the case answers). Each metric's labels sum to 100%.

Environment configuration	Masked correctness				Factual consistency			Environment cost and time	
	same category (incl. exact)†	different category	not comparable	no label*	consistent	inconsistent	no label*	Total cost (USD)	Average answer time (s)
1. Gemini 3 Flash knowledge, no search (gemini-3-flash-preview) set A / B	**86.5%**	13.5%	0.0%	0.0%	96.7%	0.0%	3.3%	$0.660	72.8
2. Current with changes 1–4 (Europe PMC + LitSense) set A / B	82.3%	17.0%	0.0%	0.7%	97.1%	0.4%	2.6%	$0.416	94.3
3. OpenRouter search: Exa instant, DeepSeek v4 Flash set A / B	85.8%	13.5%	0.7%	0.0%	96.0%	0.0%	4.0%	$3.258	74.2
4. OpenRouter search: Parallel basic, DeepSeek v4 Flash set A / B	80.9%	19.1%	0.0%	0.0%	96.3%	0.4%	3.3%	$2.516	84.2
5. OpenRouter search: Perplexity, DeepSeek v4 Flash set A / B	84.4%	14.9%	0.7%	0.0%	**97.4%**	0.0%	2.6%	$2.470	78.8
6. OpenRouter search: Google (native), Gemini 3.1 Flash Lite set A / B	83.0%	16.3%	0.7%	0.0%	96.7%	0.0%	3.3%	$21.086	82.4
7. OpenRouter search: OpenAI (native), GPT-6-luna set A / B	85.1%	14.9%	0.0%	0.0%	95.6%	0.0%	4.4%	$10.800	79.9
8. Case information only, no retrieval (DeepSeek v4 Flash 0731) set C	99.6%	0.4%	0.0%	0.0%	100.0%	0.0%	0.0%	$0.026	7.7

- Judge panel: google/gemini-3.5-flash-lite, qwen/qwen3.8-flash, openai/gpt-6-luna; each answer's label is the majority vote (with no majority, the first judge decides). Judging cost $3.30 in total, not included in the costs above.
- † Same category (incl. exact): answers at least in the true value's category (exact + same category). With different category, not comparable, and no label it sums to 100%.
- * No label: medsim gave no answer (almost all are questions its query builder refuses), or the run failed; it is not a judge label.
- Total cost: every environment component for all questions of the configuration (medsim Stages A–C, search fees, the search or knowledge model's tokens), from OpenRouter's usage.cost. Average answer time: mean wall time per question, all components included.
- When the documents give no usable basis, or there are none, medsim's synthesizer makes a best guess from the case and diagnosis.
- Configuration 1 has no search: google/gemini-3-flash-preview gets the question, the case information as medsim sees it, and the diagnosis, and gives the value that typically fits. Configurations 6 and 7 use the model provider's own search; their documents are the search model's quotations of the sources it cites. Configuration 8 has retrieval switched off.